

Chapter 116 — Retropubic Suspension Surgery for Incontinence in Females

DEFINITIONS, PREVALENCE, AND COST

- **Urinary incontinence (UI)** — International Continence Society (ICS): the involuntary loss of urine.
- **Stress urinary incontinence (SUI)** — involuntary loss on effort (sneezing, coughing, exertion); may also be a symptom of pelvic organ prolapse (POP).
- **Very common** (25–75% of females): over half of US females >20 years have UI — **SUI alone largest** (~26%), then mixed (MUI, ~16%), then urgency (UUI, ~10%).
- **Under-presentation:** only 25–45% seek care; ~10% of diagnosed SUI reach surgery.
- **Rising burden:** prevalence projected to rise ~55% (to ~28 million US females) and procedures ~47%, 2010–2050.
- **Risk factors:** advancing age, parity, vaginal delivery, obesity, diabetes, smoking, family history, physical activity, hormone replacement therapy.
- **SUI at any age:** about half of gym-attending females report UI; in nulliparous women, obesity, weight gain and low fitness predicted it; in nulliparous women only **body mass index (BMI) >25 and age >45 years** raised odds of pure SUI.
- **Economic cost:** substantial ambulatory and out-of-pocket costs; severe UI → high willingness to pay for continence.

Concept — Prevalence far outstrips treatment: *most affected females never present and few reach surgery, so demand for durable and acceptable (including mesh-free) operations is set to grow.*

GENERAL CONSIDERATIONS

Continence Mechanism

- **Continence needs four elements:** functioning striated muscle, well-vascularised submucosa, intact vaginal support, intact central and peripheral nervous system; loss of any → reduced outlet resistance → SUI.
- **Two mechanisms of SUI**, alone or combined: **intrinsic sphincter deficiency (ISD)** versus **urethral hypermobility** from failed support. **No test can apportion the contribution of each.**

THEORIES OF THE CONTINENCE MECHANISM

Theory	Proposed mechanism	Consequence
Pressure transmission (Enhorning)	Urethra descends out of its retropubic position on straining; pressure differential between retropubic and extrapelvic positions causes leakage	Basis of retropubic suspension
Rotation and descent	Combined rotation and descent of urethra away from retropubic position causes leakage	Basis of retropubic suspension

Theory	Proposed mechanism	Consequence
Hammock hypothesis (DeLancey, 1994)	Position is not the key: urethra is compressed against a backstop of levator ani, anterior vaginal wall, pelvic fascia and pubourethral ligament	Continence depends on a firm backstop

- **No consensus on mechanism → no single operation suits all.** Suspensions depend on urethral mobility; the artificial urinary sphincter (AUS) is included only because it is placed retropubically.

Concept — Suspension corrects hypermobility, not the sphincter: *it restores urethral position and support, so it presupposes a mobile urethra; where the defect is intrinsic, a device that compresses rather than supports (the AUS) is needed.*

Retropubic Suspension Procedures

- **Aim:** suspend and stabilise the anterior vaginal wall so that the **bladder neck and proximal urethra are held retropubically**; hence most beneficial in **SUI with urethral hypermobility**.

EVOLUTION OF RETROPUBIC AND RELATED PROCEDURES

Date	Eponym	Contribution
1864	Baker Brown	Suburethral tissue dissection and plication
1914	Kelly	Anterior repair
1920	Stoeckel	Suburethral suture plication
1949 (some date to 1944)	Marshall–Marchetti–Krantz (MMK)	Urethrovesical junction suspended to pubic periosteum either side of the symphysis
1959	Pereyra (later Stamey, Raz, Gittes)	Needle bladder-neck suspension , vaginal or abdominal approach
1961	Burch	Bladder neck to Cooper's ligament , 2–4 sutures each side
1976	Tanagho	Modern Burch: paravaginal sutures more lateral and tied loosely
1976–1989	Richardson; Shull and Baden	Paravaginal defect repair: lateral vaginal fascia to arcus tendineus / obturator internus fascia
1978, 1991	McGuire; Blaivas and Jacobs	Autologous fascial sling (slings covered in Chapter 118)
1990	Webster and Kreder	Vagino-obturator shelf (VOS): urethrovaginal tissue to obturator shelf, not Cooper's ligament
1991	Vancaillie and Schuessler	Laparoscopic colposuspension

- **Needle suspension:** needle passed **blindly** through the retropubic space; the vaginal walls support the urethra indirectly. Long accepted for hypermobility-related SUI.
- **No consensus on the best variant, but Burch has durable long-term data.**
- **Displacement by slings:** pubovaginal slings grew through the 1990s; with midurethral slings (MUS), slings were ~90% of US SUI operations by 2009 and few surgeons choose Burch first-line. **Late-2000s mesh publicity revived non-mesh options.**
- **Approach** (open, laparoscopic ± robotic, needle) follows surgeon experience, concomitant procedures, anatomy and preference.

Approaches

APPROACHES TO RETROPUBIC SUSPENSION

Approach	Access	Bottom line
Open	Pfannenstiel or midline infraumbilical incision	Burch success 85–90% at 1 year, about 70% at 5 years (Cochrane review)
Laparoscopic (from 1991)	Keyhole abdominal ports	Subjective cure equivalent to open to 18 months (systematic review of randomised trials); dearer initially but cost-effective long term on quality-adjusted life years (QALY)
Robotic-assisted laparoscopic (RAL)	As laparoscopic	High success at short follow-up
Mini-incisional Burch	1.5–2.5 cm transverse suprapubic incision; suture-passing device; direct vision plus proprioception	High short-term cure in a small cohort
Needle suspension	Needle and suture passed alongside urethra to elevate paravaginal tissue	Lower cure than open — suture pull-through and vaginal tissue tearing → abandoned

Concept — Access changes, the anchoring principle does not: *minimally invasive routes match open efficacy, whereas needle suspension failed because its hold on vaginal tissue tore, not because of its access.*

Artificial Urinary Sphincter

- **History:** Foley (1947) — air-inflatable periurethral cuff with removable pump; Scott (1972) — first commercial AUS (American Medical Systems). Now the **gold standard for post-prostatectomy SUI in males**.
- **Female role — ISD:** reproduces physiological sphincter function.

GUIDELINE POSITIONS ON THE AUS IN FEMALE SUI

Body	Position
ICS	Second-line
European Association of Urology (EAU)	Second-line
American Urological Association (AUA)	Second-line
National Institute for Health and Care Excellence (NICE, 2019)	Option for recurrent SUI after primary treatment

- **Routes:** open suprapubic, transvaginal, laparoscopic or robotic; **transvaginal disfavoured** — higher infection risk and theoretical vaginal erosion of the cuff.

PATIENT SELECTION AND CONSIDERATIONS IN TREATMENT CHOICES

- **Burch declined as MUS rose** (late 1990s–2000s) but remains viable in the settings listed under Indications.

Symptoms

- **Primary indication: SUI.** The AUA / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) guideline lists Burch alongside **MUS, autologous fascial pubovaginal sling and urethral bulking agents**.
- **MUI:** may be offered to treat the stress component; history establishes the most bothersome component.
- **Not indicated in UUI without SUI.**

History

- **Elements:** age, bother, amount, triggers and frequency of leakage, goals, prior treatment, medical, obstetric and surgical history (neurological disease, multiparity). Quantifying bother sets goals and expectations.
- **Triggers separate UUI, urgency- or stress-predominant MUI and pure SUI:** cough, sneeze, lift, walk, run → ~three-in-four probability of SUI, but a **negative history does not exclude it**.

Physical Examination

- **History alone is insufficient.** Height, weight and pelvic examination: demonstrable SUI, urethral mobility, tissue integrity, concomitant POP, oestrogen status.
- **Demonstrating SUI:** cough stress test and/or Valsalva supine → if negative → repeat **standing with a comfortably full bladder**.
- **Urethral mobility:** by observation or the less-used **cotton-swab (Q-tip) test** — angle change from rest to strain $>30^\circ$ = **hypermobility**. AUA/SUFU no longer recommends it for diagnosing SUI (a meta-analysis found it unlikely to aid diagnosis).
- **POP** staged by the Pelvic Organ Prolapse Quantification (POP-Q) system; note atrophy.
- Focused **abdominal** (prior incisions alter the plan) and **neurological** examination, and overall **performance status** before surgery.

Adjunctive Testing

- **Minimum in the index patient (AUA/SUFU): post-void residual (PVR) and urinalysis.** Further options: questionnaires, voiding diary, pad tests, urodynamics (UDS).

Questionnaires. **Validated instruments assess both UUI and SUI** and help separate them; particularly useful in comparative studies.

Voiding diary. **One to three days** of intake and voiding pattern; helps when symptoms are unclear, guides diagnosis and management, and improves the patient's own quantification of bother.

Pad weight tests. **Pads per day is not a pad weight test:** the weight test confirms urine loss and quantifies volume (pad weight before versus after).

PAD WEIGHT TESTS — SHORT-TERM VERSUS LONG-TERM

	Short-term	Long-term
Duration	20 minutes – 2 hours	12–72 hours, normal activity, with voiding diary
Purpose	Confirms UI	Quantifies UI
Thresholds	>1.4 g positive; less = perspiration or vaginal discharge	Per 24 h: 4.4–20 g mild; 21–74 g moderate; >75 g severe

- **ICS standardised 1-hour test:** drink 500 mL sodium-free fluid over 15 minutes (or retrograde fill to 150–300 mL, or 50–75% of capacity) → walk and climb stairs → stand from sitting $\times 10$ → cough $\times 10$ → run on the spot 1 minute → bend over $\times 5$ → wash hands under running water 1 minute → reweigh pad.

Urodynamics.

- **Indication:** history suggests SUI but no objective leakage on examination or pad test.
- Record urethral function as **Valsalva leak point pressure (VLPP)** or **maximal urethral closure pressure (MUCP): VLPP <60 cm H₂O or MUCP <20 cm H₂O suggests ISD** — significance contested.
- **For the Burch, urethral mobility matters more than urethral function.** Pressure-flow study and video add information on function and patency.

- **AUA/SUFU: UDS not required in the index patient if SUI is demonstrated on examination.**

Predictors of failure. A matched long-term cohort (Burch followed about 10 years, MUS about 13 years) found **similar long-term failure, about one in five for each**, with different predictors:

PREDICTORS OF LONG-TERM FAILURE — BURCH VERSUS MUS

Predictor	Burch	MUS
BMI	>25	>30
Preoperative anticholinergic use	Yes	Yes
Prior SUI procedure	Yes	Yes
Age >60 years	Yes	—
Smoking; diabetes	—	Yes

- **Pattern:** shared predictors (anticholinergic use, prior surgery) flag urgency and scarred tissue; the Burch fails at a **lower BMI threshold and with age**, the MUS with **tissue-healing factors** (smoking, diabetes). The same group later summarised the highest risks as high BMI, MUI and prior anti-incontinence surgery.

Patient Goals

- **Arguably the most important selection factor.** Conservative (behavioural modification, pelvic floor physiotherapy ± biofeedback, pessary) through to surgery; choice follows **bother and quality-of-life (QoL) impact**, with accurate expectations.
- **Objective and subjective outcomes correlate poorly.** Patients with a "good" baseline QoL on the Incontinence Impact Questionnaire (IIQ) should expect **smaller QoL gains** than those starting "moderate" or "poor"; goal-attainment work supports **defining pretreatment goals**.
- **Discuss mesh preoperatively:** most patients know of it from the media; re-education raises acceptance. Complicated patients who did not recall counselling were **less satisfied** → **document mesh-complication counselling**. Burch and autologous slings suit those declining mesh.

Concept — Selection turns on two questions — is the urethra mobile, and what does the patient want?

Mobility decides whether suspension can work; goals (mesh aversion, concomitant surgery, tolerance of risk) decide whether it should be chosen.

SPECIFIC DISCUSSION REGARDING RETROPUBIC SUSPENSIONS

Pertinent Anatomy

- **Retropubic space (space of Retzius):** avascular, preperitoneal potential space. Anterior — pubic symphysis; lateral — pubic rami and obturator internus; posterior — bladder; floor — endopelvic fascia (fibrofatty outer vaginal wall), paravesical fascia and levator ani fibres. Function: **support of urethra and bladder**.
- **Pubourethral ligaments:** at the proximal urethra, fanning from the lower pubis.
- **Arcus tendineus fascia pelvis (ATFP):** posterior symphysis → along the obturator internus margin → **ischial spine**. Vaginal attachments here give support; weakness → **paravaginal defect**.
- **Iliococcygeus** (levator ani) arises from the ATFP and passes to the pelvic floor midline.

Indications

- **Assess:** age, health, prior surgery, hypermobility, sphincter status, parity, fertility plans, concomitant surgery (POP repair, hysterectomy).

- **Favourable settings: concomitant open, laparoscopic or robotic abdominopelvic surgery; difficult vaginal access** (leg contractures, vaginal stenosis) favouring retropubic over sling; **reluctance to have mesh**.

CONTRAINDICATIONS AND RISK FACTORS FOR FAILURE

Precluding (contraindications)	Increase failure risk (not absolute)
Lack of vaginal mobility	Baseline urgency
No urethral mobility — negative cotton-swab test gave about five-fold higher failure → sling or AUS instead	Obesity; menopause
ISD as the sole cause of SUI — poor candidate	Prior hysterectomy; prior SUI surgery; advanced POP

- **AUS:** endorsed as **second-line**; best evidence of benefit in ISD, which arises in two settings — **recurrent SUI after prior anti-incontinence treatment** and **neurogenic SUI**. Extreme form: the "**lead-pipe urethra**" — immobile, fixed, rigid and open.

Concept — Indications are reasons to avoid the vagina or mesh; contraindications are an absent substrate for suspension — *without a mobile urethra and vagina there is nothing to lift, so the defect must be treated by a sling or sphincter instead.*

Retropubic Surgical Techniques

- **Four mainstays** (Burch, MMK, paravaginal, VOS) differ in **anchoring point and role of the sutures**; **all elevate the bladder neck, so all need urethral mobility** — a well-supported urethra with presumed ISD should respond less (speculative).
- **No concomitant abdominal surgery → do not choose a retropubic operation**, given successful vaginal options that avoid an abdominal incision.

Perioperative Considerations

Positioning and preparation. **Low lithotomy or supine with gentle leg abduction** (vaginal and retropubic access together) → prepare from at or above the umbilicus to the perineum, including the vagina → **sterile Foley catheter** (drains bladder; identifies urethra and bladder neck) → optional bladder filling to outline bladder, bladder neck and proximal urethra.

Surgical Technique

Burch Colposuspension

Open Burch. Suspends the anterior vaginal wall and para-bladder-neck tissue to **Cooper's (iliopectineal) ligament**. Exposure:

1. Pfannenstiel or lower midline incision → recti separated in the midline → anterior pelvic reflection separated from the bladder.
2. Optional **Cherney incision** (rectus tendon divided transversely above the pubis with diathermy, reflected cephalad) for access.
3. Fat and veins swept cranially and laterally to free bladder and urethra.
4. Bladder neck, proximal urethra and paraurethral vagina identified by **palpating the Foley balloon** ($\pm$ retrograde filling, intravaginal finger or end-to-end anastomosis (EEA) bowel sizer); prior-surgery scar divided sharply off the pubis.
5. Bladder dissected off the vagina **starting laterally**; mobility checked with two vaginal fingers.

6. Fat cleared to expose both Cooper's ligaments — white tendinous condensation over the periosteum of the superior pubic ramus.
- **Stay lateral — midline dissection over the urethra** risks venous-plexus bleeding, urethrovesical injury, entry into tract or vagina, and **obstruction if sutures lie too close to the bladder neck**.

BOX 116.1 DANGER POINTS AND SPECIAL CONSIDERATIONS DURING EXPOSURE FOR RETROPUBIC SUSPENSION

- **Do not dissect over the urethra** or in the midline over the bladder neck.
- **Preserve detrusor and urethral muscle.**
- Vagina and urethrovesical tissue are very vascular — identify and avoid **Santorini's plexus**.
- When passing sutures into Cooper's ligament, **be ready to control sudden, copious bleeding**.

Suture placement. Sutures **elevate the bladder neck and induce scarring and fibrosis** that maintains the elevation.

1. **No. 0 or 1-0 delayed-absorbable or non-absorbable suture on an SH tapered needle.**
2. Thimble finger or small EEA sizer in the vagina identifies and elevates the vagina, confirms wall mobility and **protects the urethra**.
3. First suture at the **level of the mid-urethra, as lateral as possible**, through vaginal wall **without entering the vaginal lumen**; contralateral suture likewise.
4. More cephalad sutures **progressively more lateral**, at the bladder neck / urethrovesical junction; **all passed medial to lateral**.
5. Paravaginal sutures passed through **Cooper's ligament** (or ATRF); heavy bleeding → pressure, clips or sutures.
6. **Distal sutures tied first, knots above (anterior to) the ligament**, with vaginal elevation.
7. Confirm **space between bladder neck and pubis — about two fingerbreadths**.
- **Modification:** paravaginal tissue elevated to the internal obturator fascia instead.

Laparoscopic or robotic Burch. Same suspension to Cooper's ligament (first described 1991); **intraperitoneal or extraperitoneal**.

Marshall-Marchetti-Krantz Procedure

- **Bladder neck suspended to the periosteum of the pubis** either side of the symphysis, replacing the urethrovesical junction in an intra-abdominal (retropubic) position; positioning and entry as for Burch.
- **Two pairs of sutures through the full thickness of the vaginal wall** either side of proximal urethra / bladder neck → periosteum; buttresses the paraurethral region. Modification: include the anterolateral urethral wall.

Paravaginal Repair

- Repairs the **fascial weakness where pelvic fascia attaches to obturator fascia / arcus tendineus**: paravaginal fascia and lateral vaginal sulcus attached to the **ATRF**, incorporating obturator fascia and muscle **from pubic symphysis to ischial spine** → restores the urethrovesical angle.

Vagino-Obturator Shelf

- **Hybrid of Burch and paravaginal repair:** laterally placed vaginal sutures elevate the vagina to the **internal obturator fascia**.
- 1. Enter the retropubic space → expose anterior bladder wall at the bladder neck (Foley balloon).
- 2. Intravaginal finger elevates the vaginal wall at bladder-neck level.

3. Sweep endopelvic fascia off the inferolateral bladder neck and proximal urethra to expose obturator muscle and ATRP.
4. Approximate endopelvic fascia and vagina to obturator muscle bilaterally → ± reinforce with a suture through the iliopectineal ligament.

Technical Considerations

- Suture placement governs elevation; anchoring point governs durability.

THE FOUR RETROPUBIC SUSPENSIONS COMPARED

Technique	Anchor	What holds the repair	Urethrovesical angle	Obstructive tendency
Burch	Cooper's ligament	Sutures	High anchor → V-configuration	Greater
MMK	Pubic periosteum at symphysis	Sutures	High anchor → V-configuration	Greater
Paravaginal	ATRP, symphysis to ischial spine	Tissue approximation and scarring	Lower, flatter	Less
VOS	Internal obturator fascia (± iliopectineal ligament)	Tissue approximation and scarring	Lower, flatter	Less

Artificial Urinary Sphincter

- Placed retropubically **at the bladder neck** in females. **Open approach traditional** because a vaginal incision raises the risk of cuff exposure; robotic placement reported in case series.

AUS PLACEMENT — OPEN SUPRAPUBIC VERSUS TRANSVAGINAL

Step	Open suprapubic	Transvaginal
Access	Suprapubic incision → preperitoneal space → space of Retzius	Broad inverted-U vaginal incision , so the closure does not overlie the cuff → flap to bladder neck
Cuff	Bladder neck dissected, measured; fitted cuff placed	Proximal urethra and bladder neck freed circumferentially, measured; fitted cuff placed
Pump	Pocket in labia majora	Labia majora pocket via the same vaginal incision
Reservoir	Preperitoneal, behind rectus abdominis	Small infraumbilical incision into preperitoneal space
Special point	—	Copious antibiotic irrigation is critical to prevent prosthetic infection

KEY POINTS: ANATOMY AND MECHANISM OF DIFFERENT APPROACHES

- **Burch**: restores urethrovesical angle and retropubic position by **broad support** of urethra and bladder neck.
- **MMK**: fixes periurethral tissue to the **pubic symphysis / periosteum**.
- **Paravaginal repair**: reattaches lateral vaginal sulcus to the **arcus tendineus fascia pelvis** from pubic symphysis to ischial spine.
- **AUS**: restores the sphincter mechanism by **circumferential compression** in females with ISD.

Perioperative Care

- Catheter at the surgeon's discretion; unnecessary if a voiding trial is passed in recovery.

- **Incomplete emptying — two strategies:** suprapubic tube placed intraoperatively (historical, for comfort) versus teaching **clean intermittent catheterisation (CIC)** — less invasive, avoids suprapubic trauma, lets the patient monitor emptying in real time.
- Either way, void and check PVR **until consistently <150 mL**.

Concept — Anchor height sets the trade-off: *high, suture-dependent anchors (Burch, MMK) lift more and angle the urethra into a V, at a cost of obstruction; low, scar-dependent anchors (paravaginal, VOS) lie flatter and obstruct less.*

OUTCOMES

Important Considerations

- **Few reproducible objective measures;** success, perspective, outcomes and tools must be defined.
- **UDS: most objective, poorly predictive.** In Burch versus autologous sling arms, **urodynamic SUI predicted success but leak point pressure did not;** normal UDS did not predict postoperative voiding dysfunction.
- **International Consultation on Incontinence (ICI):** trials should enrol by **symptom-driven criteria;** **no urethral function measure** should be the sole measure of SUI severity or predictor of surgical outcome.

Definitions of Success

- **Definitions of cure have varied widely,** hindering comparison; standardisation efforts continue.
- **Objective success: cough stress test plus 1-hour pad test.**
- **Subjective success:** one or more of Urogenital Distress Inventory (UDI/UDI-6), IIQ, King's Health Questionnaire (KHQ), ICI Questionnaire–Short Form (ICIQ-SF), Patient Global Impression of Improvement (PGI-I).

Patient Perspective

- **Expectations must match what treatment can deliver;** counselling needs the qualitative patient perspective defined and measured (e.g. Patient-Centered Outcomes Research Institute, PCORI).

Method of Follow-Up

- **Early postoperative contact (AUA/SUFU);** telemedicine is safe after MUS (inferred for suspensions), and app-based follow-up gave higher satisfaction than office visits.
- **In-person visit by 6 months:** history, examination and PVR; ask about continence, voiding, overactive bladder (OAB) symptoms, sexual function and new symptoms; questionnaires optional.

Length of Follow-Up

ACCEPTED FOLLOW-UP DURATIONS

Category	Duration
Short-term	1–2 years
Intermediate-term	12 or 24 months to 47 months
Long-term	>48 months

- **Results deteriorate with time** → distrust short-term reports. Long-term data are limited by loss to follow-up, few examinations, questionnaire-only outcomes and variable definitions.

Clinical Outcomes

Comparative Studies

- **Burch versus conservative treatment:** scant, poor-quality evidence; a subset analysis favoured Burch over pelvic floor muscle training (PFMT).
- **Open suspension versus periurethral injection:** one randomised trial (surgery pooled: Burch, needle suspension, slings) — **surgery more successful at 12 months**; collagen had **fewer and less severe complications**; no difference in satisfaction or QoL.
- **Versus other non-sling operations** (needle suspension, anterior colporrhaphy, MMK, paravaginal repair): comparators largely obsolete — see the Cochrane review.

TRIALS AND SYNTHESSES — BURCH VERSUS SLINGS

Study	Design and population	Comparison	Result	Verdict
SISTEr (Stress Incontinence Surgical Treatment Efficacy trial, Urinary Incontinence Treatment Network, 2007)	RCT; 655 randomised 1:1; 520 completed	Open Burch vs autologous fascial pubovaginal sling (PVS)	2 years: PVS more successful ; more urinary tract infection, lower urinary tract symptoms, de novo UI; lower flow and higher detrusor pressure at maximum flow — sling more obstructive (both raise outlet resistance)	PVS more effective, more morbid
SISTEr 5-year follow-up (2012)	Observational extension	As above	Continence lower after Burch ; satisfaction (tracking continence) higher after sling despite more voiding dysfunction; both declined yearly	Sling advantage durable
Ward and Hilton (2002; 2004; 2008) — first RCT of TVT vs colposuspension	Urodynamic SUI; 179 tension-free vaginal tape (TVT) vs 169 open colposuspension	TVT vs open colposuspension	No difference in objective cure (strict definition → low absolute rates; underpowered for equivalence); 6-month no leakage 36% vs 28% ; TVT: more bladder perforation; colposuspension: more voiding dysfunction , slower return to normal voiding; parity held at 2 and 5 years (1-hour pad test; Bristol Female Lower Urinary Tract Symptoms questionnaire)	Comparable
Matched cohort	Large matched cohort	Burch vs MUS	Comparable success, satisfaction, complications and ICIQ-SF scores	Comparable
RCT, two centres	Small	Laparoscopic Burch vs TVT	Both improved; more postoperative UI and urodynamic SUI after Burch; longer operating time for Burch	Favours TVT
Meta-analysis of RCTs (confirmed by later meta-analyses)	39 RCTs	Retropubic MUS vs Burch	Higher continence and cure with MUS; more bladder perforation with retropubic MUS	Favours MUS

Study	Design and population	Comparison	Result	Verdict
Retrospective, 5 years	—	Burch vs transobturator MUS	Comparable outcomes; less voiding dysfunction and retention with transobturator MUS	Comparable

- **Shared risks of Burch and slings:** de novo storage symptoms (8–27% after colposuspension; 3–23% after PVS); voiding dysfunction (2–27% after colposuspension; up to 11% after PVS, with 1.5–7.8% needing long-term CIC); **POP after colposuspension (2.5–27%)**.
- **Concomitant surgery raises complications regardless of technique** (cystitis and need for CIC, both greater after sling).
- **Pattern across the table:** slings match or beat Burch on continence; the price is obstruction and morbidity (autologous PVS) or bladder perforation (retropubic MUS), whereas the Burch carries more voiding dysfunction than TVT and later POP.

Concomitant Pelvic Organ Prolapse

- **Concomitant anti-incontinence surgery benefits females undergoing POP surgery**, whether SUI is overt or occult (meta-analyses).

BURCH AT THE TIME OF SACROCOLPOPEXY

Study	Population	Comparison	Result
CARE (Colpopexy and Urinary Reduction Efforts, 2006; 2-year data 2008)	322 stress-continent females having abdominal sacrocolpopexy	Prophylactic Burch (157) vs none (165)	Less SUI severity and bother with Burch , sustained at 2 years
Secondary analysis of an RCT	Females with preoperative SUI having sacrocolpopexy	MUS vs Burch	MUS superior for stress-specific continence at 1 and 2 years; no difference in POP recurrence, voiding dysfunction, urgency or satisfaction

- **The contrast:** in stress-continent females, adding a Burch is better than nothing; in females who already have SUI, MUS beats Burch.

Artificial Urinary Sphincter

- **Evidence base thin** — randomised controlled trials (RCTs) called for. **Meta-analysis:** most patients continent, a minority improved; **explantation, revision, vaginal erosion and mechanical failure** substantial; complications — infection, mechanical malfunction, **urethral erosion causing recurrent SUI**, explantation. **Surgeon experience is decisive** given technical complexity.
- **Minimally invasive versus open (meta-analysis):** overall fewer complications with laparoscopic or robotic placement — driven by much **less mechanical failure** and less cuff erosion — although infection and atrophy were somewhat higher. **Erosion rises with follow-up beyond 5 years.**
- **Systematic reviews:** most had prior SUI surgery; **bladder neck and vaginal injuries**, widely varying continence, de novo retention; **some cannot manage the pump**. In non-neurogenic ISD with recurrent severe SUI: 0-pad continence in roughly half to most, **substantial revision and explantation** — comparable to transobturator slings in a fixed urethra.

Complications

- **Suture through urethra or bladder** → urinary tract infection, irritative voiding, stones, fistula.

- **Intraoperative cystoscopy is mandatory:** cystoscopy → confirm ureteric efflux → if absent, release the sutures → repeat until ureteric patency confirmed; also **check no suture has passed through the Foley catheter**. Bleeding concern → retropubic drain.
- **Postoperative voiding dysfunction** (OAB and/or obstruction); **de novo POP after Burch**.
- **Osteitis pubis after MMK (0.7–2.5%):** presents **2–12 weeks** postoperatively with suprapubic pain radiating to the lower limbs and a tender symphysis → may resolve spontaneously → or progress to **osteomyelitis** (antibiotics) → worst case, surgical debridement.

Other Considerations

- **Understudied:** cost comparison, postoperative sexual function, trainee exposure — **procedure volume is not keeping pace with trainee numbers**.

Concept — Burch is the durable, mesh-free benchmark that slings match or narrowly beat: *its niche is set by patient and context (concomitant abdominal surgery, mesh avoidance, no vaginal access), not by superior efficacy.*

CONCLUSIONS

- **Viable in the properly selected patient:** those undergoing an abdominal procedure and those wishing to avoid a graft, synthetic or biological.
- **Successful and durable;** the **Burch remains a guideline recommendation** for surgical treatment of female SUI.